Homework

Assume, in addition to the money market account which yields a constant return $r > 0$, there are two stocks with dynamics:

$$\frac{dS_t^{(1)}}{S_t^{(1)}} = \mu_1 dt + \sigma_1 dW_t^{(1)}; \quad \frac{dS_t^{(2)}}{S_t^{(2)}} = \mu_2 dt + \sigma_2 \left(\rho dW_t^{(1)} + \sqrt{1-\rho^2} dW_t^{(2)}\right) \tag{1}$$

where $\mu_1, \mu_2 \in \mathbb{R}$, $\sigma_1, \sigma_2 > 0$, $-1 < \rho < 1$, and $W^{(1)}, W^{(2)}$ are two independent standard Brownian motions under the physical probability measure P. As such, ρ is correlation in the instantaneous asset returns. In our portfolio we are long a European option with time T terminal payoff:

$$V(T, S_T^{(1)}, S_T^{(2)}) = \left(S_T^{(1)} - S_T^{(2)}\right)^2$$

the squared distance between the two stocks, and we are delta-hedging according to the (two dimensional) Black-Scholes delta hedging strategy.

Recall that the arbitrage free price of this option at $t \leq T$ is:

$$V(t, S_t^{(1)}, S_t^{(2)}; \sigma_t, \rho_t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} \left(S_T^{(1)} - S_T^{(2)}\right)^2 \,\middle|\, S_t^{(1)}, S_t^{(2)} \right]$$

Here $\mathbb{Q}$ is the risk neutral measure under which each risk asset has drift rate r. Above, by "σ_t" and "ρ_t" we mean that we have estimated the volatility and correlation at time t to be σ_t, ρ_t respectively and then to price the option, we assume the volatility and correlation are kept at σ_t, ρ_t over $[t, T]$. While this is not entirely consistent with (0.1), it is common industry practice.

We are using the standard delta-hedging strategy. As such, our position at time t is 1 unit notional of the option, and $-h_t^{(1)}, -h_t^{(2)}$ shares of the stocks, where:

$$h_t^{(1)} = \partial_{s_1} V(t, s_1, s_2; \sigma_t, \rho_t); \quad h_t^{(2)} = \partial_{s_2} V(t, s_1, s_2; \sigma_t, \rho_t)$$

The portfolio value at time t is thus:

$$V_t = V(t, S_t^{(1)}, S_t^{(2)}; \sigma_t, \rho_t) - h_t^{(1)} S_t^{(1)} - h_t^{(2)} S_t^{(2)}$$

The horizon is 5 days, where one day is $\Delta = 1/252$. We hold a constant share position over $[t, t + 5\Delta]$ so that at $t + 5\Delta$ the value of our portfolio is:

$$V_{t+5\Delta} = V(t + 5\Delta, S_{t+5\Delta}^{(1)}, S_{t+5\Delta}^{(2)}; \sigma_{t+5\Delta}, \rho_{t+5\Delta}) - h_t^{(1)} S_{t+5\Delta}^{(1)} - h_t^{(2)} S_{t+5\Delta}^{(2)}$$

Again, by $\sigma_{t+5\Delta}, \rho_{t+5\Delta}$ we mean the constant volatility and correlation, now estimated at $t + 5\Delta$ which we use to price the option over $[t + 5\Delta, T]$. We are interested in running a scenario analysis to see what our losses look like if at the end of 5 days the log returns $X_{t+5\Delta}^{(i)} = \log(S_{t+5\Delta}^{(i)}/S_t^{(i)})$, $i = 1, 2$; volatility $\sigma_{t+5\Delta}$; and correlation $\rho_{t+5\Delta}$ takes certain values.

The scenarios under consideration are any combination of:

$$\begin{aligned} X_{t+5\Delta}^{(i)} &= \pm 0.20, \pm 0.10, \pm 0.05; && i = 1, 2 \\ \frac{\sigma_{t+5\Delta}^{(i)}}{\sigma_t^{(i)}} &= 0.5, 0.75, 1.25, 1.5, 1.75, 2; && i = 1, 2 \\ \rho_{t+5\Delta} - \rho_t &= 0, \pm 0.5 \end{aligned}$$

Thus, there are $N = 3888$ possible scenarios $x_1, \ldots, x_N$. For each scenario x_n we will compute the full portfolio loss $l_n := L_{t+5\Delta}(x_n) = -(V_{t+5\Delta}(x_n) - V_t)$.

Problems

(a) Show that:

$$V(t, s_1, s_2; \sigma, \rho) = e^{r(T-t)} \left(s_1^2 e^{\sigma_1^2 (T-t)} - 2 s_1 s_2 e^{\sigma_1 \sigma_2 \rho (T-t)} + s_2^2 e^{\sigma_2^2 (T-t)} \right)$$

(b) Output the worst case scenario risk measure:

$$\rho(L_{t+5\Delta}) = \max\{l_n \mid n = 1, \ldots, N\}$$

along with the log return/volatility combination which achieves this measure.

(c) Now, assume the following weights ($i = 1, 2$):

$$X_{t+5\Delta}^{(i)} = \begin{cases} \pm 0.20 & \text{weight} = 0.5 \\ \pm 0.10 & \text{weight} = 0.75 \\ \pm 0.05 & \text{weight} = 1 \end{cases}$$

$$\frac{\sigma_{t+5\Delta}^{(i)}}{\sigma_t^{(i)}} = \begin{cases} 0.5, 2 & \text{weight} = 0.5 \\ 0.75, 1.75 & \text{weight} = 1.25 \\ 1.25, 1.5 & \text{weight} = 0.75 \end{cases}$$

$$\rho_{t+5\Delta} - \rho_t = \begin{cases} 0 & \text{weight} = 0.5 \\ \pm 0.5 & \text{weight} = 1.25 \end{cases}$$

so that, for example, the weight of $X_{t+5\Delta}^{(1)} = -0.1$, $X_{t+5\Delta}^{(2)} = 0.2$, $\sigma_{t+5\Delta}^{(1)} = 2\sigma_t^{(1)}$, $\sigma_{t+5\Delta}^{(2)} = 0.5\sigma_t^{(2)}$ and $\rho_{t+5\Delta} - \rho_t = -0.5$ is $w = 0.75 \times 0.5 \times 0.5 \times 0.5 \times 1.25 = 15/128$. Compute the scenario risk measure:

$$\rho(L_{t+5\Delta}) = \max\{w_n l_n \mid n = 1, \ldots, N\}$$

along with the log return/volatility/correlation combination which achieves this measure. For this combination, what is the portfolio loss?

For parameter values to use:

$$r = 1.32\%, \quad \mu_1 = 15.475\%, \quad \mu_2 = 18.312\%$$

$$t = 0: \quad \sigma_{1,t} = 22.14\%, \quad \sigma_{2,t} = 30.36\%, \quad \rho_t = 0.237$$

$$T = 0.25, \quad S_0^{(1)} = 158.12, \quad S_0^{(2)} = 170.33, \quad \Delta = 1/252$$